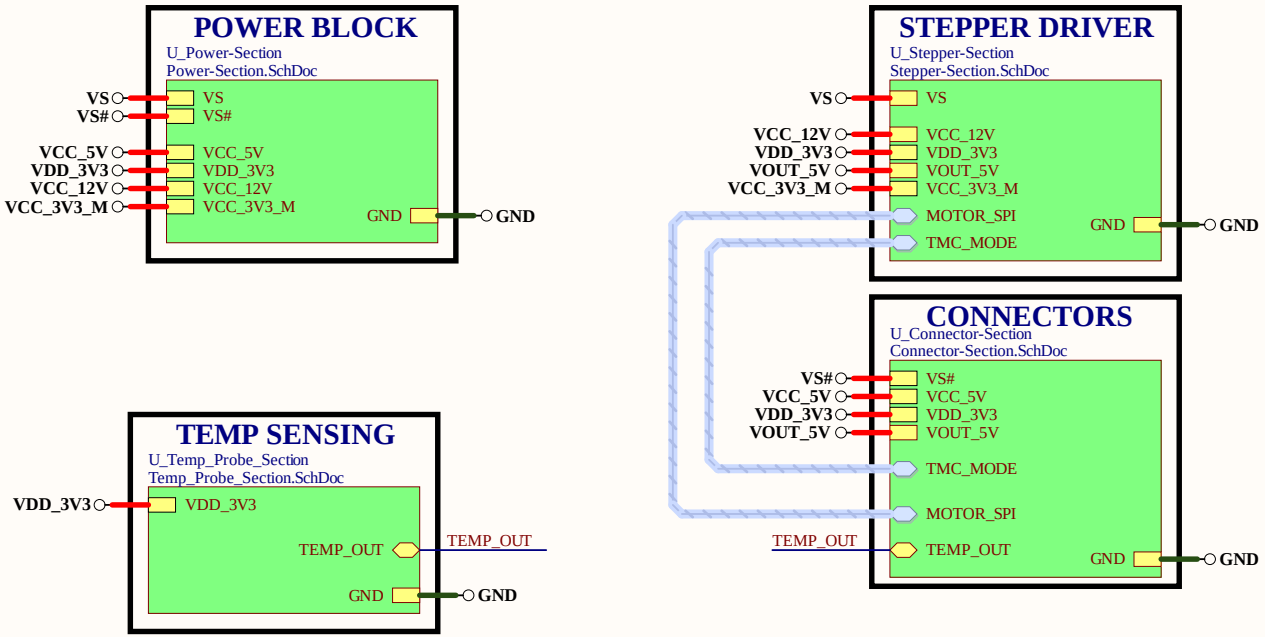
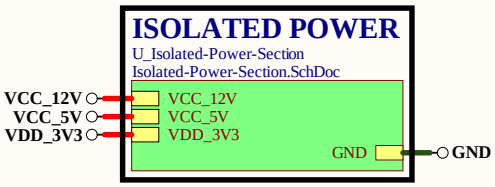
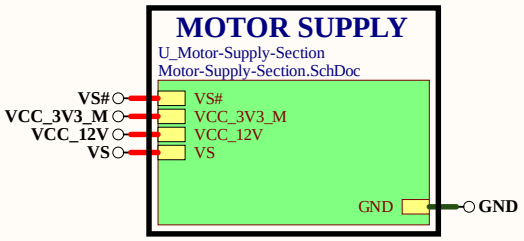
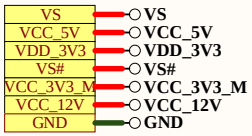
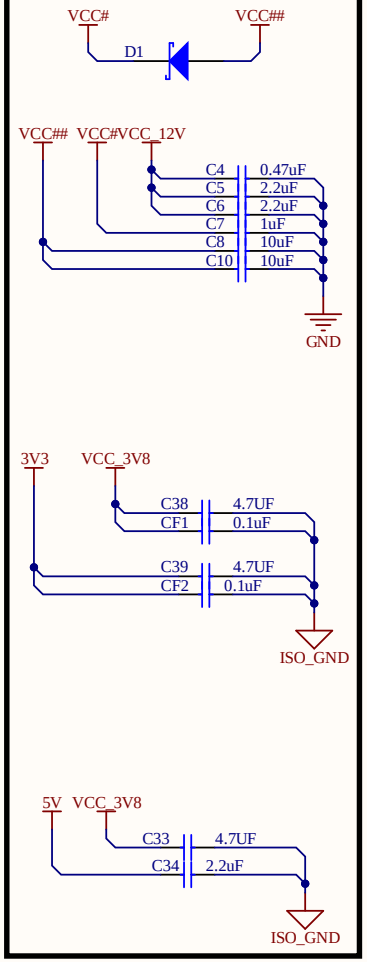

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VS#
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
VS
VCC_12V
VDD_3V3
VOUT_5V
VCC_3V3_M
MOTOR_SPI
TMC_MODE
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VS#
VCC_5V
VDD_3V3
VOUT_5V
TMC_MODE
MOTOR_SPI
TEMP_OUT
GND
GND
TEMP SENSING
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
TEMP_OUT
GND
GND


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
VCC_12V
VCC_5V
VDD_3V3
GND
GND

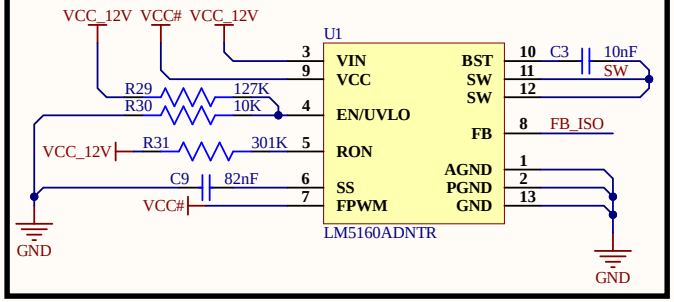

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VS#
VCC_3V3_M
VCC_12V
VS
VS#
VCC_3V3_M
VCC_12V
VS
GND
GND


VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND
VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND

DECOUPLING CAPS

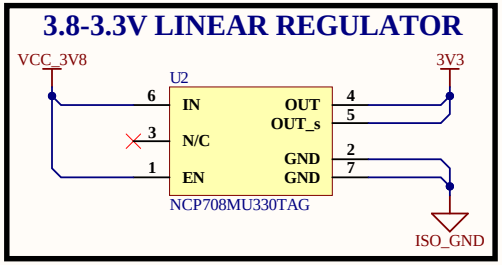


12V-3.8V ISOLATED BUCK CONVERTER

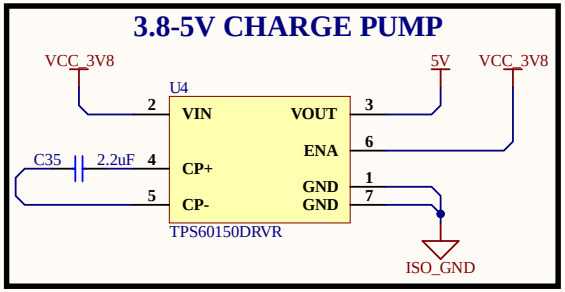


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

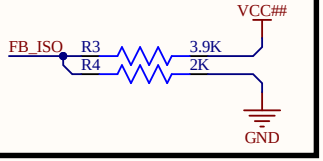
3.8-3.3V LINEAR REGULATOR



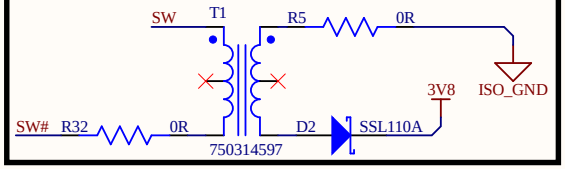
3.8-5V CHARGE PUMP



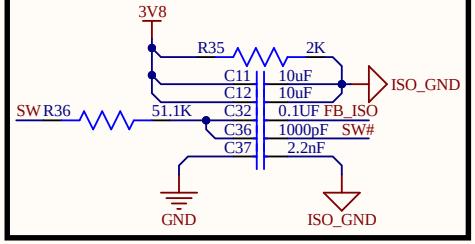
FEEDBACK NETWORK



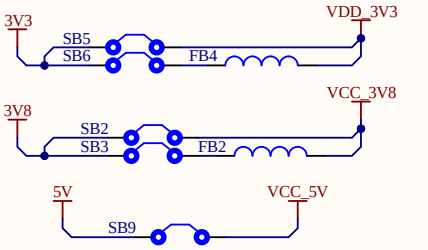
FLYBACK TRANSFORMER



SUPPORTING CIRCUIT

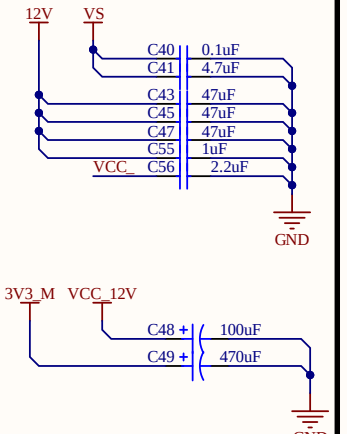


SOLDER BRIDGE CONNECTIONS

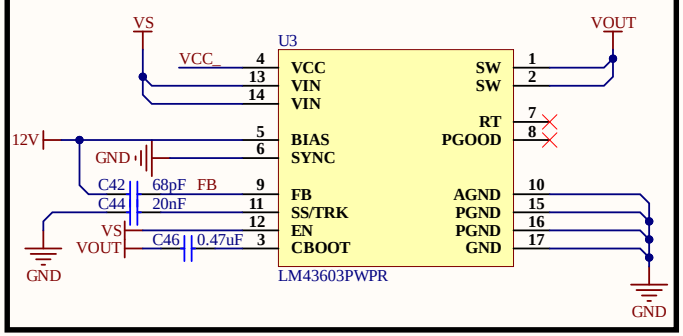


VCC_12V — VCC_12V
VCC_5V — VCC_5V
VDD_3V3 — VDD_3V3
GND — GND

DECOUPLING CAPS



30V-12V NON-ISOLATED BUCK CONVERTER

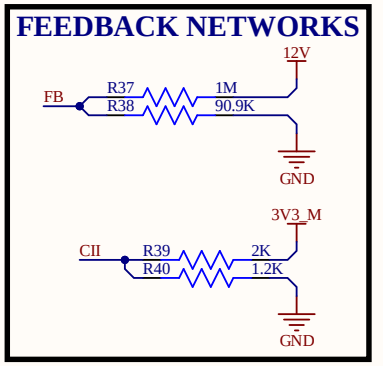


For VOUT=12V, RFBB= 90.9K, RFBT=1M (Datasheet pg:26)

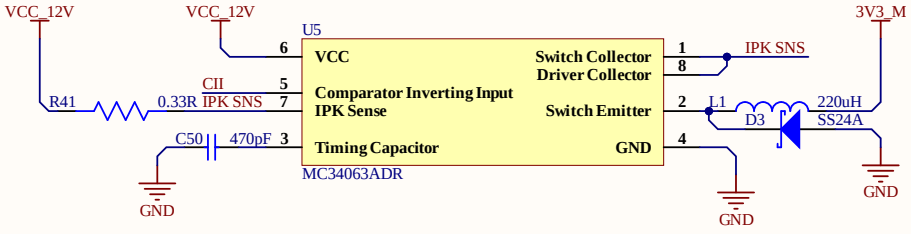
INDUCTOR



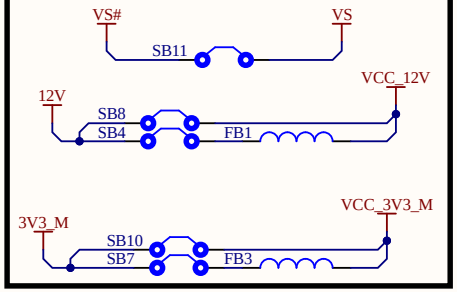
FEEDBACK NETWORKS

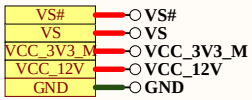


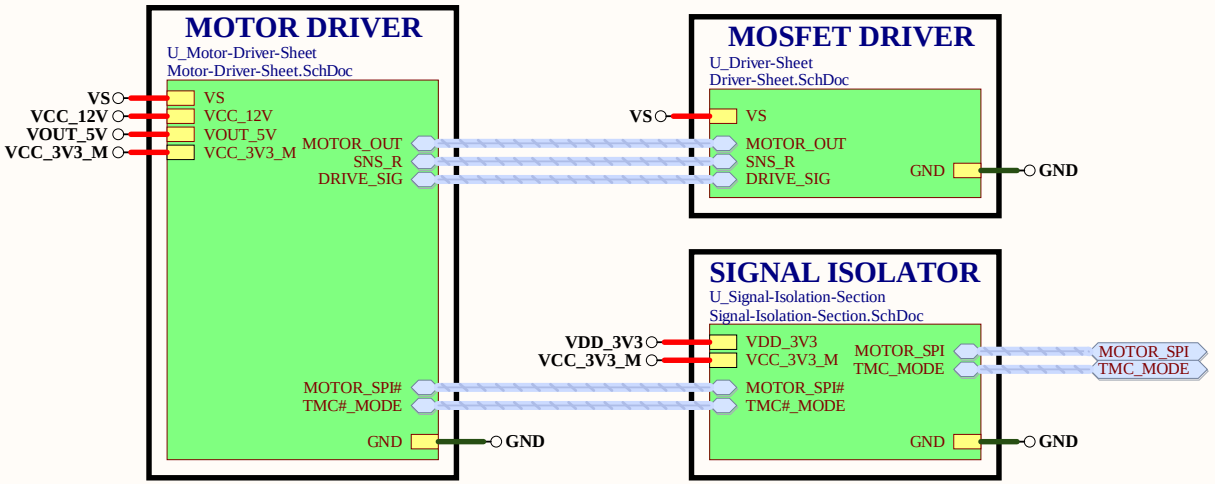
12-3.3V BUCK CONVERTER

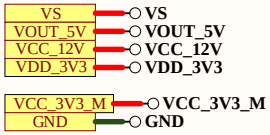


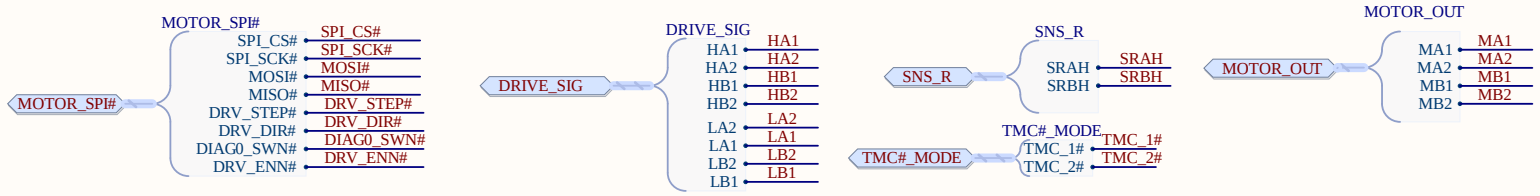
SOLDER BRIDGE



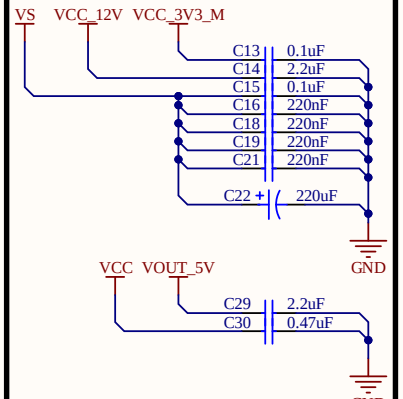



MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
MOTOR_SPI
TMC_MODE

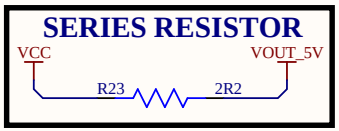

VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND



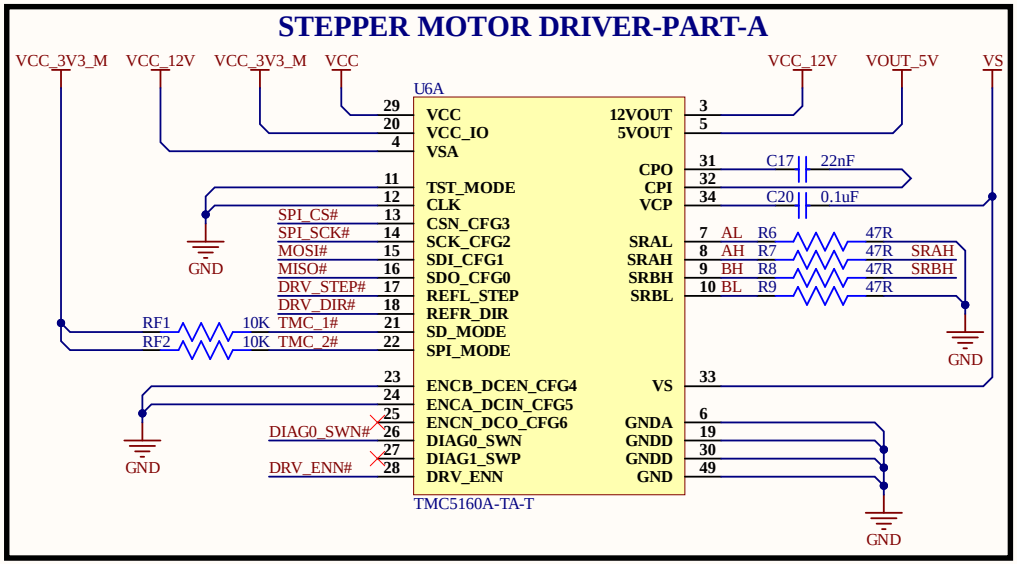
DECOUPLING CAPS



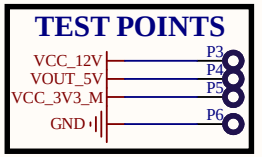
SERIES RESISTOR



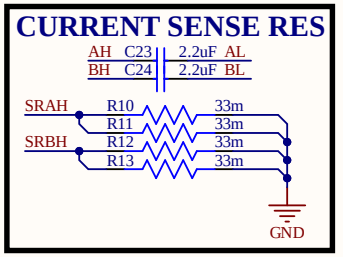
STEPPER MOTOR DRIVER-PART-A



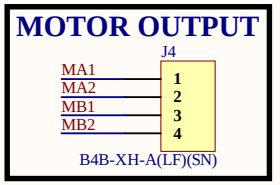
TEST POINTS



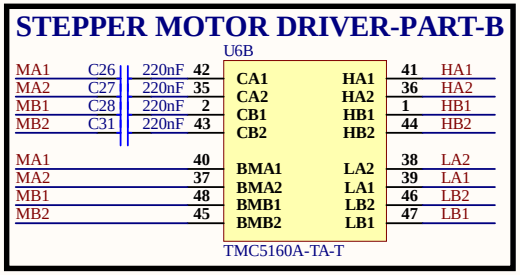
CURRENT SENSE RES

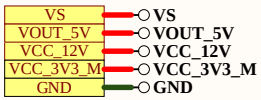


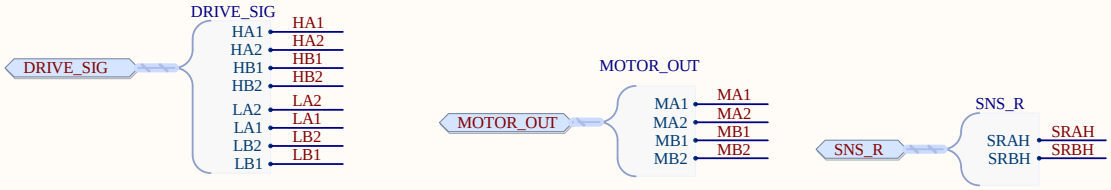
MOTOR OUTPUT



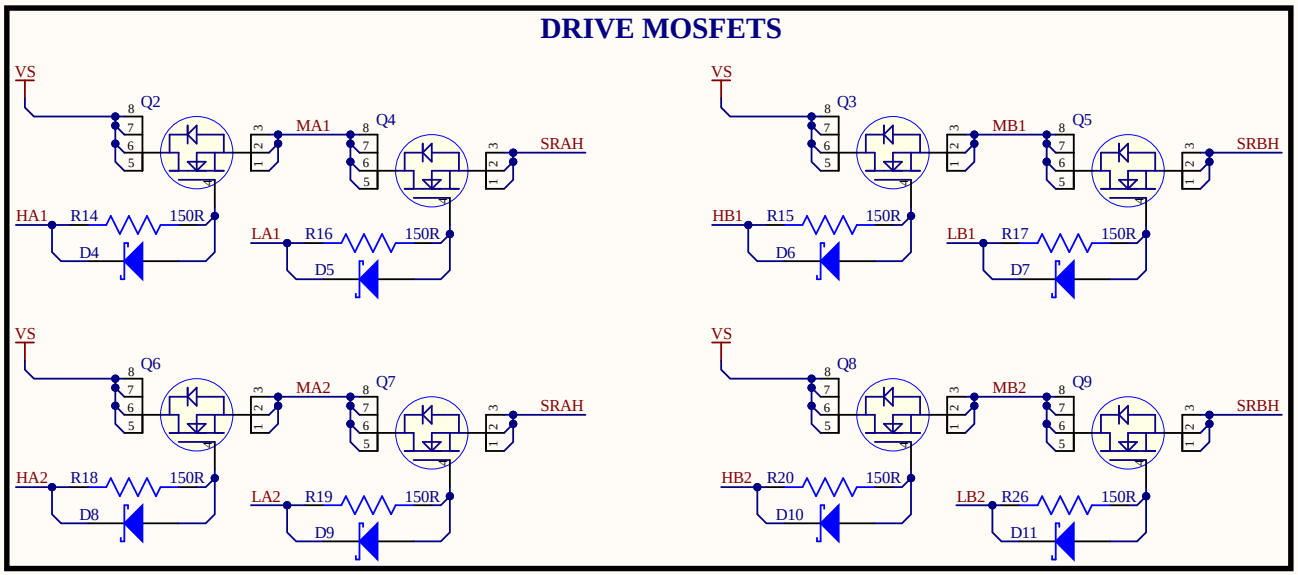
STEPPER MOTOR DRIVER-PART-B



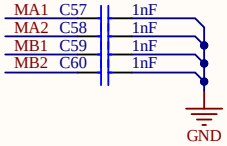




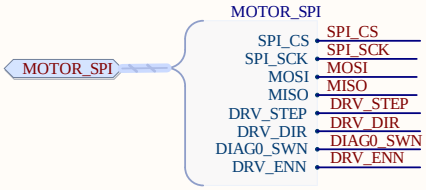
DRIVE MOSFETS

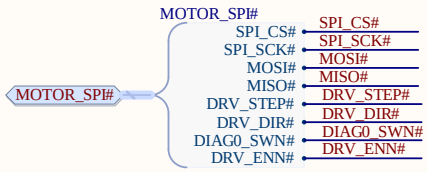


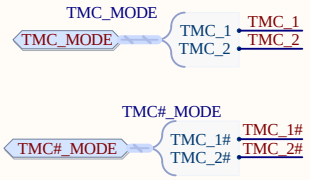
DECOUPLING CAPS

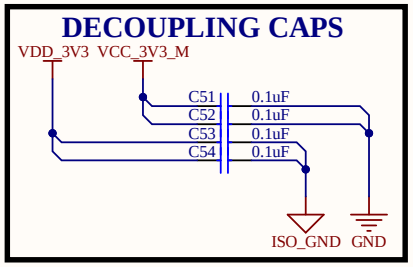


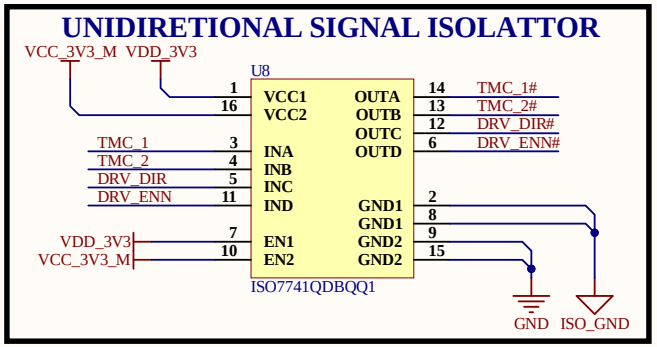


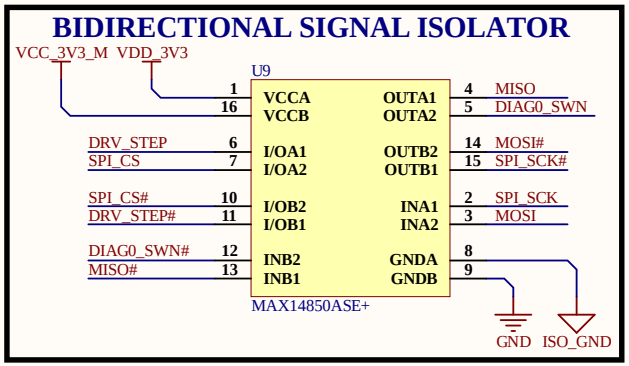










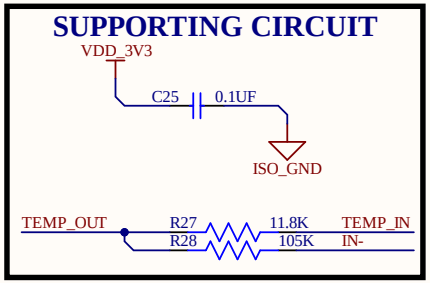




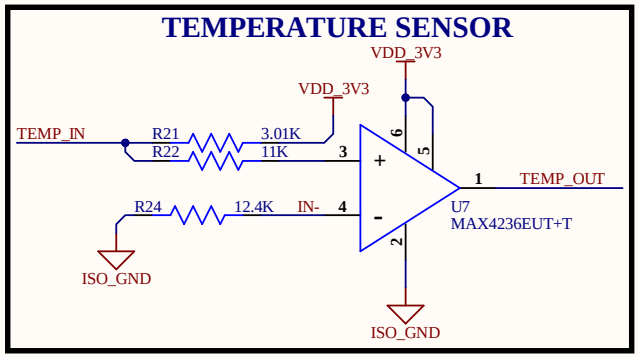
Title		
Size: A4	Number	Revision
Date: 3/17/2022	Sheet of	
File: D:\Github\..\Signal-Isolation-Section.SchDoc	Drawn By:	

TEMP_OUT TEMP_OUT

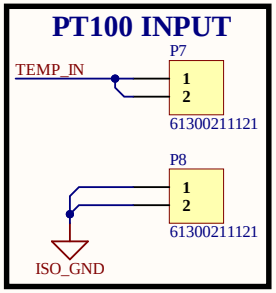
SUPPORTING CIRCUIT



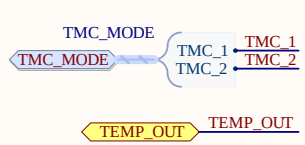
TEMPERATURE SENSOR

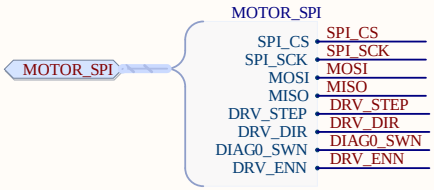


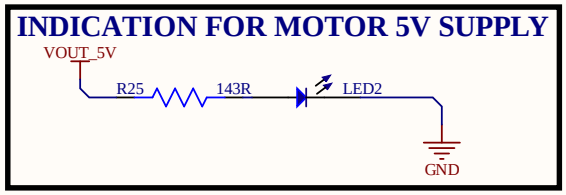
PT100 INPUT

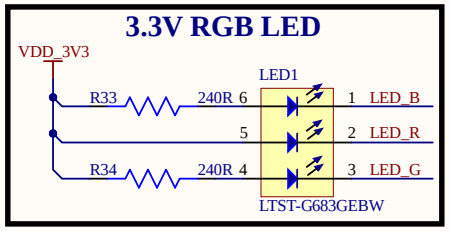


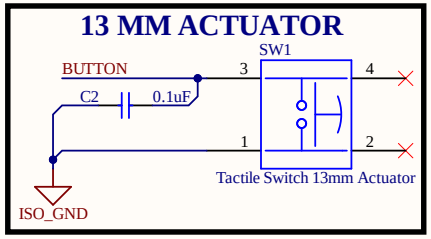
VDD_3V3 **VDD_3V3**
GND **GND**

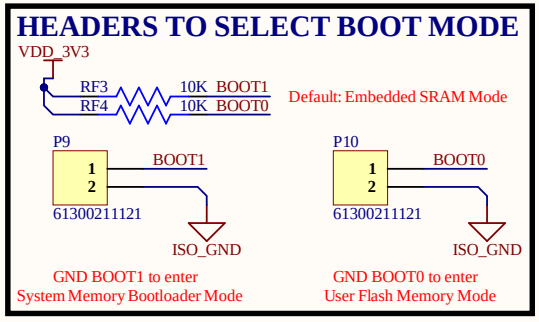

TMC_MODE
TMC_MODE
TMC_1
TMC_2
TMC_1
TMC_2
TEMP_OUT
TEMP_OUT

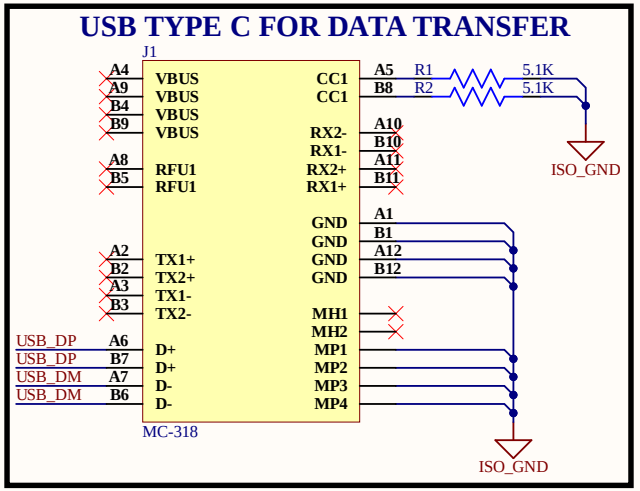

MOTOR_SPI
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN

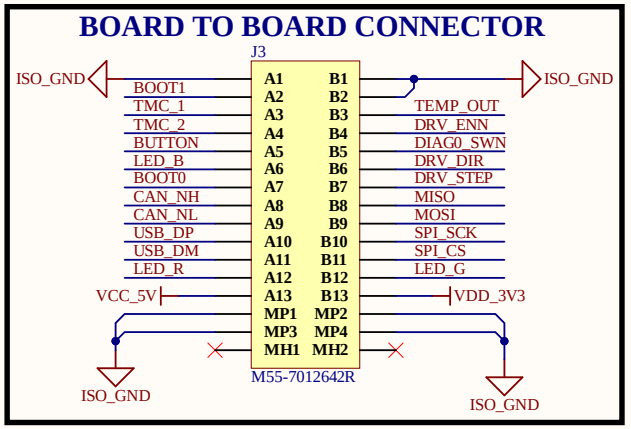

INDICATION FOR MOTOR 5V SUPPLY
VOUT_5V
R25
143R
LED2
GND

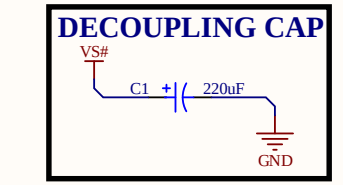

3.3V RGB LED
VDD_3V3
LED1
R33
240R
6
1
LED_B
5
2
LED_R
R34
240R
4
3
LED_G
LTST-G683GEBW

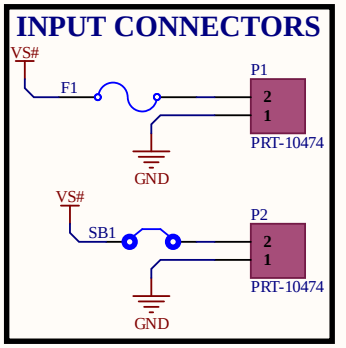

13 MM ACTUATOR
SW1
BUTTON
3
4
C2
0.1uF
1
2
Tactile Switch 13mm Actuator
ISO_GND

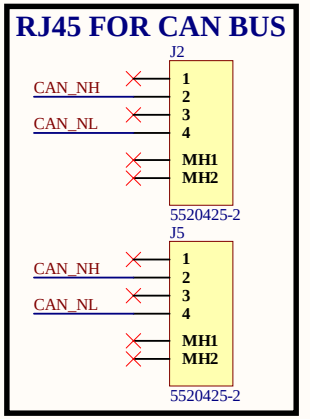

HEADERS TO SELECT BOOT MODE
VDD_3V3
RF3
10K
BOOT1
RF4
10K
BOOT0
Default: Embedded SRAM Mode
P9
1
2
BOOT1
61300211121
ISO_GND
P10
1
2
BOOT0
61300211121
ISO_GND
GND BOOT1 to enter
System Memory Bootloader Mode
GND BOOT0 to enter
User Flash Memory Mode

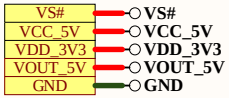

USB TYPE C FOR DATA TRANSFER
J1
A4
A9
B4
B9
VBUS
A8
B5
RFU1
A2
B2
A3
B3
TX1+
TX2+
TX1-
TX2-
USB_DP
A6
B7
USB_DP
USB_DM
A7
B6
USB_DM
D+
D-
CC1
CC1
RX2-
RX1-
RX2+
RX1+
GND
A5
B8
R1
R2
5.1K
5.1K
ISO_GND
A10
B10
A11
B11
A1
B1
A12
B12
MH1
MH2
MP1
MP2
MP3
MP4
MC-318
ISO_GND


BOARD TO BOARD CONNECTOR
J3
ISO_GND
ISO_GND
BOOT1
TMC_1
TMC_2
BUTTON
LED_B
BOOT0
CAN_NH
CAN_NL
USB_DP
USB_DM
LED_R
VCC_5V
TEMP_OUT
DRV_ENN
DIAG0_SWN
DRV_DIR
DRV_STEP
MISO
MOSI
SPI_SCK
SPI_CS
LED_G
VDD_3V3
M55-7012642R
ISO_GND
ISO_GND


DECOUPLING CAP
VS#
C1
220uF
GND


INPUT CONNECTORS
VS#
F1
P1
2
1
PRT-10474
GND
VS#
SB1
P2
2
1
PRT-10474
GND


RJ45 FOR CAN BUS
J2
CAN_NH
CAN_NL
1
2
3
4
MH1
MH2
5520425-2
J5
CAN_NH
CAN_NL
1
2
3
4
MH1
MH2
5520425-2


VS#
VCC_5V
VDD_3V3
VOUT_5V
GND
VS#
VCC_5V
VDD_3V3
VOUT_5V
GND